SAMSUNG

Streamlining Global Manufacturing: SAP Production Planning (PP) at Samsung

A strategic overview of enterprise resource planning in consumer electronics manufacturing

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Driving the Tech Supply Chain

Samsung Electronics stands as a global manufacturing powerhouse, producing millions of devices—from smartphones to semiconductors—across interconnected facilities worldwide.

SAP Production Planning (PP) serves as the central nervous system for Samsung's manufacturing operations, orchestrating three critical decision pillars:

- **WHAT:** Strategic device allocation and product mix.
- **WHEN:** Precise scheduling for flagship global launches.
- **HOW MUCH:** Optimized inventory levels to meet volatile market demand.

Global Organizational Structure

The SAP PP organizational hierarchy maps Samsung's manufacturing footprint into logical, scalable units that streamline planning and execution.

Client

Samsung Group

The overarching global environment housing all business operations and enterprise data.

Company Code

Samsung Electronics Co. Ltd.

The primary unit for maintaining autonomous legal financial records and books.

Plant and Storage Organization

We structure our manufacturing sites into distinct physical and logical units to optimize production capacity and inventory management across the global supply chain.

Plant

Defined as the primary production site, each plant (e.g., Suwon, Noida) operates with unique capabilities, specific capacity constraints, and independent master data.

Storage Location

These units allow for precise stock segmentation, enabling the clear separation of raw components like microchips from finished goods awaiting distribution.

Core Master Data: The Digital Recipe

Precision is the cornerstone of our manufacturing excellence. Every Galaxy device is born from a rigorous digital master data framework—a sophisticated blueprint that translates complex design specifications into high-performance reality across our global production network.

Material Master: The Source of Truth

This foundational database codifies every physical component, from high-resolution OLED panels to advanced mobile processors. It establishes standardized unit measurements, technical specifications, and logistical storage parameters, ensuring total transparency across our inventory ecosystem.

Bill of Materials (BOM): The Structural Blueprint

The BOM acts as the master orchestration layer, mapping the precise assembly sequence of 200+ distinct components. It synchronizes complex supply chains to ensure that every finished Galaxy is constructed with flawless consistency, regardless of where in the world it is produced.

By unifying Material Master data with precise BOM structures, we eliminate operational ambiguity, minimize production variance, and accelerate time-to-market for our latest mobile innovations.

Core Master Data: Routings & Work Centers

With the ingredients defined, we now codify the manufacturing process—the precise "culinary technique" that dictates how these components are assembled into a finished Galaxy.

Routing

The definitive operational sequence: Motherboard assembly, screen integration, battery installation, quality validation, and final packaging.

Work Centers

The specialized hubs—assembly lines and robotic cells—where production occurs, tracking capacity, cycle times, and resource availability.

Sales and Operations Planning (SOP)

Demand Forecasting

SAP PP leverages machine learning to synthesize historical sales, seasonal cycles, and market shifts into highly accurate demand models. By transforming complex datasets into actionable foresight, we ensure market-aligned production.

Master Production Scheduling (MPS)

MPS translates demand into a concrete production roadmap. For instance, anticipating a 15% Q4 holiday growth triggers automated procurement and capacity scaling six months early, ensuring seamless fulfillment during peak demand.

Material Requirement Planning (MRP)

As the backbone of Samsung's production agility, the SAP PP MRP engine autonomously orchestrates complex net requirement calculations to ensure **zero-stockout operations** for critical, high-precision components.

01

Demand Aggregation

Translates production targets into gross component requirements—such as the precise materials needed for a **1-million-unit** smartphone batch.

02

Inventory Reconciliation

Performs real-time availability checks by cross-referencing required quantities against existing stock levels of microchips, displays, and batteries.

03

Net Shortage Calculation

Identifies precise gaps: if **1 million** chips are needed but only **800,000** are available, the system instantly isolates a **200,000-unit** deficit.

04

Automated Procurement

Triggers immediate purchase requisitions and supplier orders to bridge identified shortages, keeping the supply chain perfectly synchronized.

Manufacturing Execution on the Shop Floor

The shop floor acts as the physical realization of our digital strategy. By bridging the gap between planning and assembly, we ensure seamless production flow, real-time resource allocation, and absolute inventory visibility from the first order to the final product shipment.

1. Planned Order

MPS-driven production roadmap.

2. Production Order

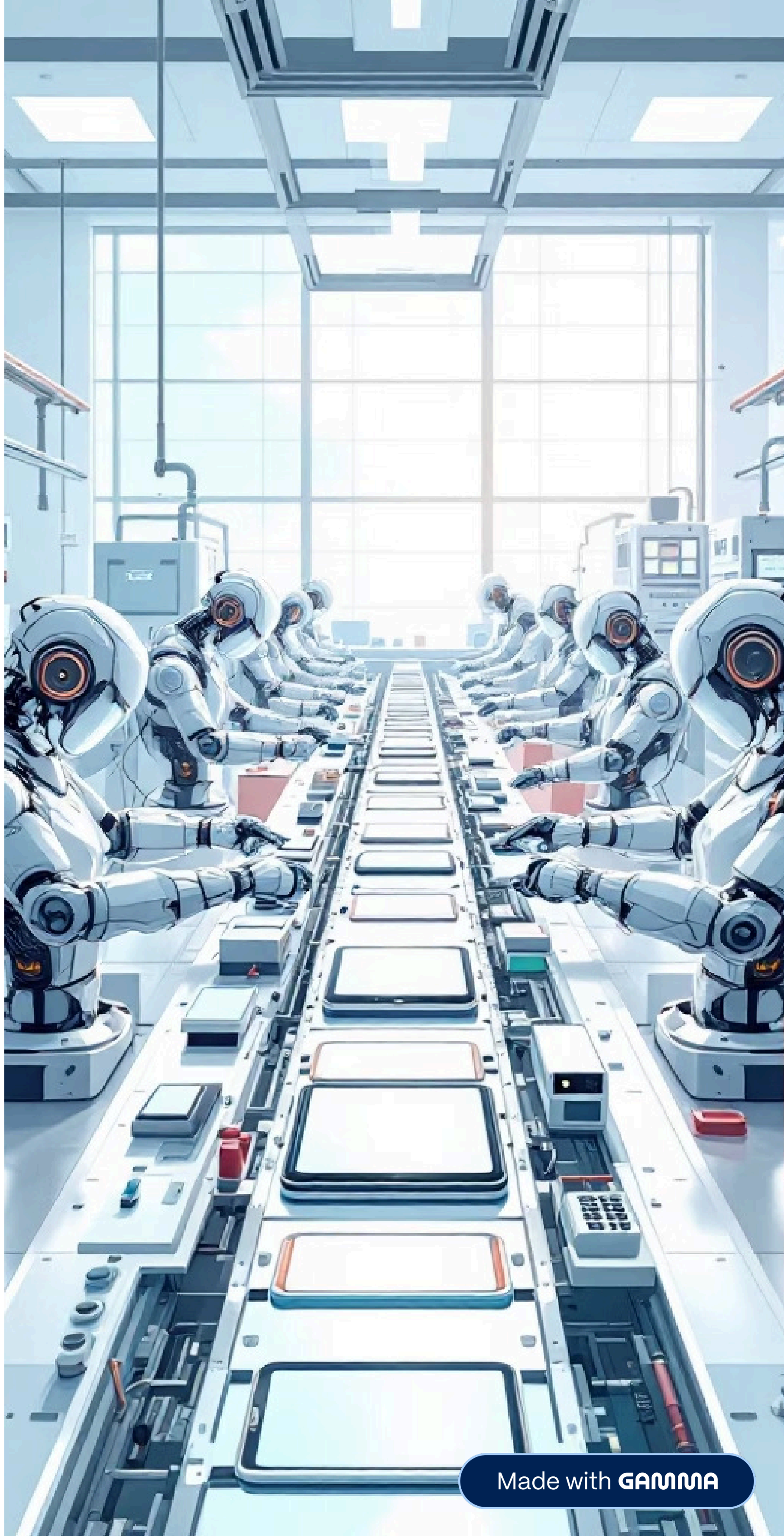
Execution-ready demand with confirmed resource allocation.

3. Material Withdrawal

Precision staging of components from warehouse to line.

4. Goods Receipt

Final quality confirmation and warehouse entry.



Strategic Impact of SAP PP

SAP PP serves as the digital backbone for Samsung's manufacturing dominance, enabling operational excellence at a global scale.

Global Scalability

Unifying production across 100+ facilities with standardized quality and compliance.

Inventory Precision

Optimizing working capital by eliminating stockouts and minimizing excess carrying costs.

Agile Responsiveness

Real-time supply chain visibility enabling instant adaptation to volatile market demand.

Strategic Alignment

Synchronizing capacity, materials, and delivery to meet complex production mandates.